

Minh Quang Bui

Mechatronics Engineering Student

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Education

Ontario Tech University

BEng (Honours) Mechatronics Engineering

June 2028

Oshawa, ON

Relevant Coursework: Mechanical Design, Thermodynamics, Fluid Mechanics, Control Systems

Experience

Moist Sanding Operator — Assembly (TPT)

Oct 2025 – Present

General Motors Canada

Oshawa, ON

- Perform surface finishing and inspection on automotive components in high volume environment
- Identify surface defects and support root cause analysis to reduce rework and quality issues
- Follow standardized work, GM Global Manufacturing System (GMS), and 5S safety practices
- Collaborate with **production and maintenance** teams to sustain uptime and workflow efficiency
- Maintain clean, organized workstation to ensure consistent quality output

Automotive Assembly & Mechanical Technician Intern

Jan 2024 – Aug 2025

JAC Motors (Vietnam) & St. Mary's HS

- Supported mechanical assembly and servicing using hand and power tools in manufacturing environment
- Interpreted **technical drawings** to ensure correct **assembly** and component fitment
- Performed preventive maintenance including fluid service, battery replacement, and engine work
- Operated **drill press, milling machines, and vehicle lifts** following strict safety procedures
- Assisted troubleshooting mechanical issues to improve system reliability and reduce downtime

Projects

Smart Garage Door Automation

Jul 2024 – Dec 2024

Arduino, C/C++, Control Theory, SolidWorks

- Developed **embedded control** logic for automated electromechanical system with safety limits
- Integrated **sensors**, motors, and actuators with real-time system response and debugging
- Performed functional testing and validation to ensure reliable system operation
- Created 3D **CAD models** and assemblies in **SolidWorks** for mechanical design and integration

Robotics Engineering Lead

Feb 2023 – May 2024

Skills Ontario

Etobicoke, ON

- Designed mechanical chassis and assemblies using **AutoCAD** and **SolidWorks** for competition robot
- Developed PCB layouts and integrated electronic systems (SMD/through-hole design)
- Programmed **embedded C/C++** control for motors, sensors, and Bluetooth communication
- Improved mechanical system efficiency and durability through iterative testing and redesign
- Achieved **30% space reduction** and increased system reliability through design optimization

Technical Skills

CAD & Design: AutoCAD, SolidWorks (3D modeling, assemblies, technical drawings)

Manufacturing: GD&T (basic), DFMA, machining fundamentals, 3D printing

Controls & Systems: Arduino, embedded C/C++, PLC fundamentals (basic)

Mechanical: Assembly/disassembly, mechanical systems, preventive maintenance

Tools: Drill press, milling/shaping machines, hoists, lifts, hand/power tools

Manufacturing Systems: GMS, 5S, standardized work, quality inspection

Analysis: MATLAB (basic), troubleshooting, root cause analysis